MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR,
LOW-POWER SCHOTTKY TTL, DECODERS,
MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, decoder microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Single 3 to 8 line decoder
02	Dual 2 to 4 line decoder
03	BCD-to-decimal decoder
04	BCD-to-seven segment decoder/driver (15-volt, open collector output)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
X	C-2A (20-terminal, .350" x .350"), square chip carrier package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA dc to +5.5 V dc
Storage temperature range	-65° to +150°C
Maximum power dissipation (P_D): 1/	
Device type 01	55 mW dc
Device type 02	60.5 mW dc
Device types 03 and 04	71.5 mW dc
Output voltage (off state) (device type 04)	15 V dc
Output current (device type 04)	12 mA dc
Lead temperature (soldering 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases E, F, X, and 2	(See MIL-M-38510, appendix C)
Junction temperature	+175°C 2/

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.7 V dc
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. Case outlines shall be in accordance with MIL-M-38510 and 1.2.3 herein.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. Electrical test requirements shall be as specified in table II. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance, by device class, are specified in table II.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits		Unit
					Min	Max	
High-level output voltage	V_{OH1}	V_{IL} = 0.7 V V_{CC} = 4.5 V V_{IH} = 2.0 V	I_{OH} = -50 µA	04	2.4		V
High-level output voltage	V_{OH2}		I_{OH} = -400 µA	01,02,03	2.5		V
Low-level output voltage at BI/RBO	V_{OL1}		I_{OL} = 1.6 mA	04		0.40	V
Low-level output voltage	V_{OL2}		I_{OL} = 1/	01,02, 03, 04		0.40	V
Input clamp voltage	V_{IC}		I_{IN} = -18 mA, T_C = +25°C	All		-1.5	V
Maximum collector cut-off current	I_{CEX}		V_{OH} = 15 V	04		250	µA
Low-level input current at A, B, C, D, G	I_{IL1}	V_{CC} = 5.5 V V_{IN} = 0.4 V		01	-.001	-.38	mA
				02, 03	-.03	-.40	mA
				04	-.10	-.34	mA
Low-level input current at LT and RBI	I_{IL2}			04	-.11	-.36	mA
Low-level input current at BI/RBO	I_{IL3}			04	-.36	-1.37	mA
High-level input current	I_{IH1}		V_{IN} = 2.7 V	All		20	µA
	I_{IH2}		V_{IN} = 7.0 V	All		100	µA
Short circuit output current	I_{OS}		V_{IH} = 5.5 V 2/ V_{IL} = GND	01, 03	-6	-130	mA
				02	-15	-100	mA
Supply current	I_{CC}			01		10	mA
				02		11	mA
				03, 04		13	mA
Propagation delay time, low-to-high level	t_{PLH1}	V_{CC} = 5.0 V C_L = 50 pF ±10%	R_L = 665Ω ±10%	04	5	158	ns
Propagation delay time, low-to-high level through 2 levels of logic (binary select to output)	t_{PLH2}		R_L = 2.0 kΩ ±10%	01, 02	5	38	ns
Propagation delay time, low-to-high level through 2 levels of logic (A, B, C, or D to output)	t_{PLH2}			03	5	45	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits		Unit
					Min	Max	
Propagation delay time, low-to-high level through 2 levels of logic (enable to output)	t_{PLH3}	V_{CC} = 5.0 V C_L = 50 pF ±10%	R_L = 2.0 kΩ ±10%	01	5	35	ns
				02	5	44	ns
Propagation delay time, low-to-high level through 3 levels of logic (binary select to output)	t_{PLH4}			01	5	48	ns
				02	5	51	ns
Propagation delay time, low-to-high level through 3 levels of logic (A, B, C, or D to output)	t_{PLH4}			03	5	53	ns
Propagation delay time, low-to-high level through 3 levels of logic (enable to output)	t_{PLH5}			01	5	47	ns
Propagation delay time, high-to-low level	t_{PHL1}		R_L = 665Ω ±10%	04	5	158	ns
Propagation delay time, high-to-low level through 2 levels of logic (binary select to output)	t_{PHL2}		R_L = 2.0 kΩ ±10% ±10%	01	5	69	ns
				02	5	57	ns
Propagation delay time, high-to-low level through 2 levels of logic (A, B, C, or D to output)	t_{PHL2}			03	5	45	ns
Propagation delay time, high-to-low level through 2 levels of logic (enable to output)	t_{PHL3}			01, 02	5	56	ns
Propagation delay time, high-to-low level through 3 levels of logic (binary select to output)	t_{PHL4}			01	5	81	ns
				02	5	65	ns
Propagation delay time, high-to-low level through 3 levels of logic (A, B, C, or D to output)	t_{PHL4}			03	5	53	ns
Propagation delay time, high-to-low level through 3 levels of logic (enable to output)	t_{PHL5}			01	5	65	ns

1/ I_{OL} = 4 mA for device types 01, 02, and 03; I_{OL} = 12 mA for device type 04.

2/ Not more than one output should be shorted at one time.

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3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 7, 9
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 8, 9, 10, 11
Group B test requirements (method 5005), subgroup 5	1, 2, 3, 7, 8, 9, 10, 11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1, 2, 3
Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2, 3

* PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following conditional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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Pin number		Pin name		Pin name		Pin name		Pin name	
Case		Device type 01		Device type 02		Device type 03		Device type 04	
		Case		Case		Case		Case	
2, X	E, F	2, X	E, F	2, X	E, F	2, X	E, F	2, X	E, F
1	1	NC	A	NC	1G	NC	0	NC	B
2	2	A	B	1G	1A	0	1	B	C
3	3	B	C	1A	1B	1	2	C	LT
4	4	C	G2A	1B	1Y0	2	3	LT	B1/RBO
5	5	G2A	G2B	1Y0	1Y1	3	4	B1/RBO	RB1
6	6	NC	G1	NC	1Y2	NC	5	NC	D
7	7	G2B	Y7	1Y1	1Y3	4	6	RB1	A
8	8	G1	GND	1Y2	GND	5	GND	D	GND
9	9	Y7	Y6	1Y3	2Y3	6	7	A	e
10	10	GND	Y5	GND	2Y2	GND	8	GND	d
11	11	NC	Y4	NC	2Y1	NC	9	NC	c
12	12	Y6	Y3	2Y3	2Y0	7	D	e	b
13	13	Y5	Y2	2Y2	2B	8	C	d	a
14	14	Y4	Y1	2Y1	2A	9	B	c	g
15	15	Y3	Y0	2Y0	2G	D	A	b	f
16	16	NC	V_{CC}	NC	V_{CC}	NC	V_{CC}	NC	V_{CC}
17		Y2		2B		C		a	
18		Y1		2A		B		g	
19		Y0		2G		A		f	
20		V_{CC}		V_{CC}		V_{CC}		V_{CC}	

FIGURE 1. Terminal connections.

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Device type 01

INPUTS					OUTPUTS							
ENABLE		SELECT										
G1	G2*	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	H	H	H	H	H	H	H	H
H	L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	H	H	L	H	H	H	H	H	H
H	L	L	H	L	H	H	L	H	H	H	H	H
H	L	L	H	H	H	H	H	L	H	H	H	H
H	L	H	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	H	L	H	H
H	L	H	H	L	H	H	H	H	H	H	L	H
H	L	H	H	H	H	H	H	H	H	H	H	L

*G2 = G2A + G2B

H = high level, L = low level, X = irrelevant.

Device type 02

INPUTS			OUTPUTS			
ENABLE	SELECT					
G	B	A	Y0	Y1	Y2	Y3
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	L	H	H	L	H	H
L	H	L	H	H	L	H
L	H	H	H	H	H	L

H = high level, L = low level, X = irrelevant

FIGURE 2. Truth tables.

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Device type 03

INPUTS				OUTPUTS									
D	C	B	A	0	1	2	3	4	5	6	7	8	9
L	L	L	L	L	H	H	H	H	H	H	H	H	H
L	L	L	H	H	L	H	H	H	H	H	H	H	H
L	L	H	L	H	H	L	H	H	H	H	H	H	H
L	L	H	H	H	H	H	L	H	H	H	H	H	H
L	H	L	L	H	H	H	H	L	H	H	H	H	H
L	H	L	H	H	H	H	H	H	L	H	H	H	H
L	H	H	L	H	H	H	H	H	H	L	H	H	H
L	H	H	H	H	H	H	H	H	H	H	L	H	H
H	L	L	L	H	H	H	H	H	H	H	H	L	H
H	L	L	H	H	H	H	H	H	H	H	H	H	L
H	L	H	L	H	H	H	H	H	H	H	H	H	H
H	L	H	H	H	H	H	H	H	H	H	H	H	H
H	H	L	L	H	H	H	H	H	H	H	H	H	H
H	H	L	H	H	H	H	H	H	H	H	H	H	H
H	H	H	L	H	H	H	H	H	H	H	H	H	H
H	H	H	H	H	H	H	H	H	H	H	H	H	H

FIGURE 2. Truth tables - Continued.

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Device type 04

	INPUTS							OUTPUTS							
DECIMAL OR FUNCTION	LT	RBI	D	C	B	A	BI/RBO	a	b	c	d	e	f	g	NOTE
0	H	H	L	L	L	L	H	L	L	L	L	L	L	H	1
1	H	X	L	L	L	H	H	H	L	L	H	H	H	H	1
2	H	X	L	L	H	L	H	L	L	H	L	L	H	L	
3	H	X	L	L	H	H	H	L	L	L	L	H	H	L	
4	H	X	L	H	L	L	H	H	L	L	H	H	L	L	
5	H	X	L	H	L	H	H	L	H	L	L	H	L	L	
6	H	X	L	H	H	L	H	H	H	L	L	L	L	L	
7	H	X	L	H	H	H	H	L	L	L	H	H	H	H	
8	H	X	H	L	L	L	H	L	L	L	L	L	L	L	
9	H	X	H	L	L	H	H	L	L	L	H	H	L	L	
10	H	X	H	L	H	L	H	H	H	H	L	L	H	L	
11	H	X	H	L	H	H	H	H	H	L	L	H	H	L	
12	H	X	H	H	L	L	H	H	L	H	H	H	L	L	
13	H	X	H	H	L	H	H	L	H	H	L	H	L	L	
14	H	X	H	H	H	L	H	H	H	H	L	L	L	L	
15	H	X	H	H	H	H	H	H	H	H	H	H	H	H	
BI	X	X	X	X	X	X	L	H	H	H	H	H	H	H	2
RBI	H	L	L	L	L	L	L	H	H	H	H	H	H	H	3
LT	L	X	X	X	X	X	H	L	L	L	L	L	L	L	4

NOTES:
1. BI/RBO is wire-OR logic serving as blanking input (BI) and/or ripple-blanking output (RBO). The blanking input must be open or held at a high logic level when output functions 0 through 15 are desired and ripple-blanking input (RBI) must be held open or at a high logic level during the decimal 0 input, X = input may be high or low.
2. When a low logic level is applied to the blanking input (forced condition) all segment outputs go to a high logic level regardless of the state of any other input condition.
3. When ripple-blanking input (RBI) is at a low logic level, lamp test input is at high logic level and A = B = C = D = low logic level, all segment outputs go to a high logic level and the ripple-blanking output goes to a low logic level (response condition).
4. When blanking input/ripple-blanking output is open or held at a high logic level, and a low logic level is applied to lamp-test input, all segment outputs go to a low logic level.

FIGURE 2. Truth tables - Continued.

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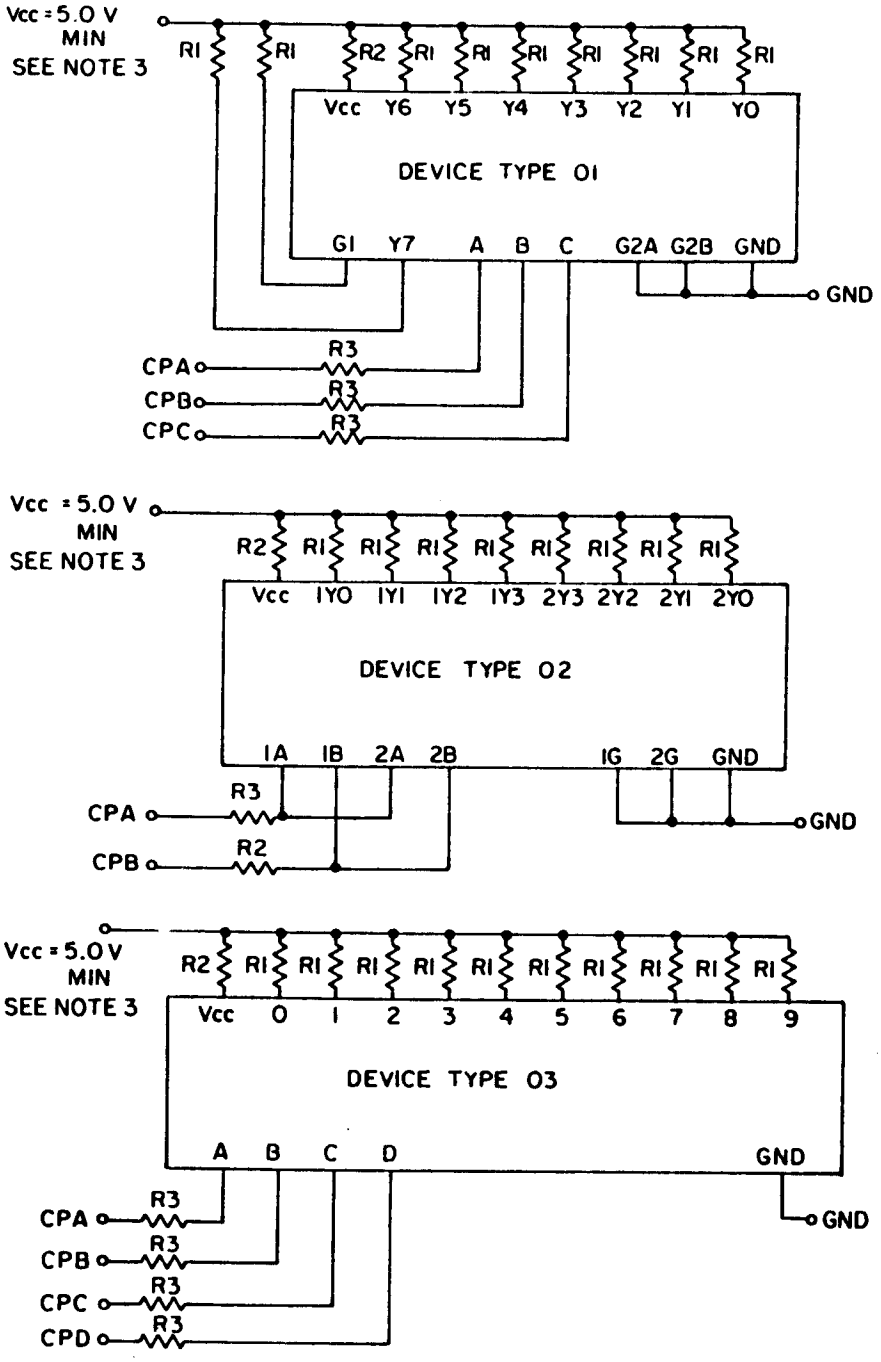
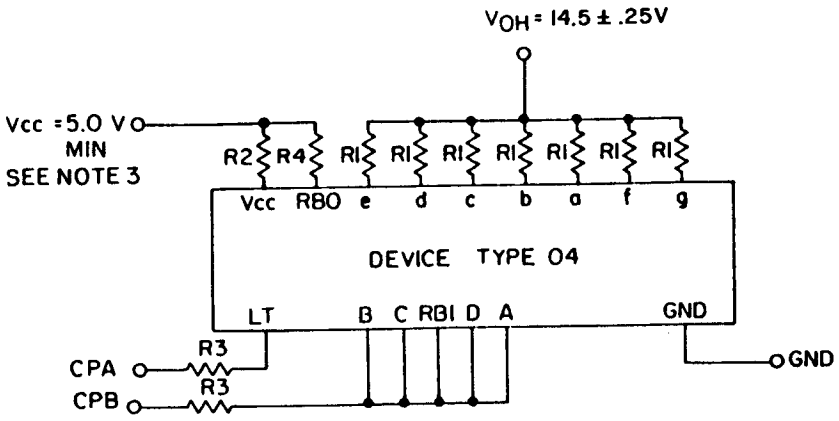
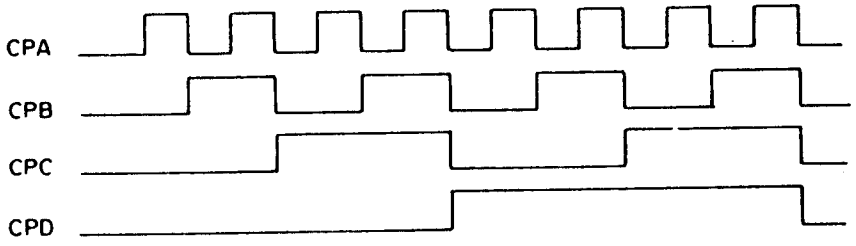


FIGURE 3. Burn-in and life test circuits.

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NOTES:
1. CPA = 100 kHz ±50% square wave; 50% duty cycle;
   V_{IL} = -0.5 V minimum to 0.7 maximum, V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CPB = Same characteristics as CPA except ½ the frequency and phase relationship shown below.
   CPC = Same characteristics as CPB except ½ the frequency and phase relationship shown below.
   CPD = Same characteristics as CPC except ½ the frequency and phase relationship shown below.



2. For device types 01, 02, 03:
   R1 = 1 kΩ ±5%; R2 = 10Ω ±5%; R3 = 27Ω ±5%.

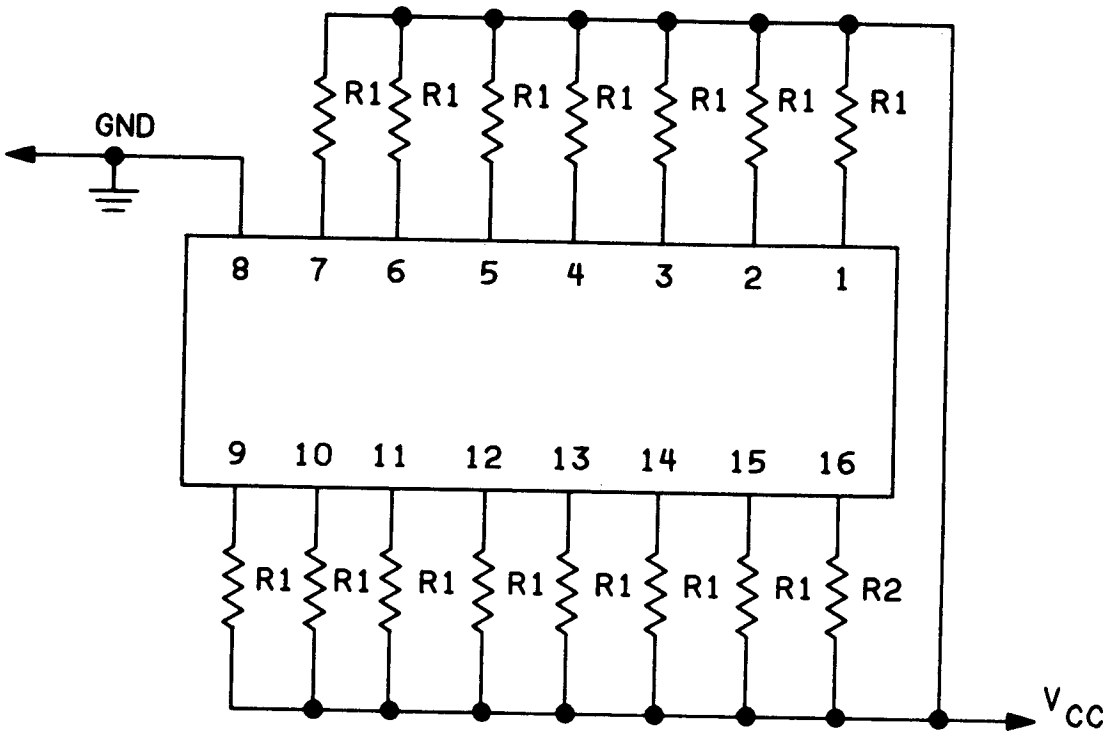
   For device type 04:
   R1 = 1300Ω ±5%; R2 = 10Ω ±5%; R3 = 27Ω ±5%; R4 = 1 kΩ ±5%.

3. V_{CC} shall be high enough to insure 5.0 V at device V_{CC} terminal.

FIGURE 3. Burn-in and life test circuits - Continued.

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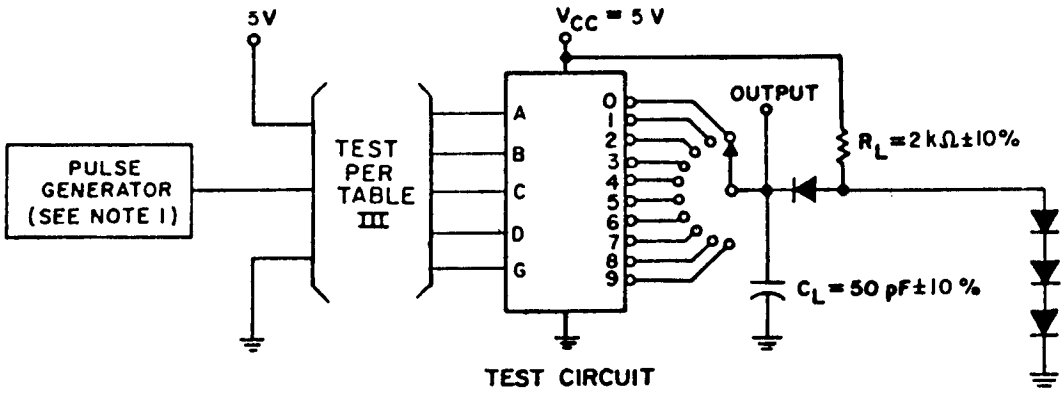
All device types



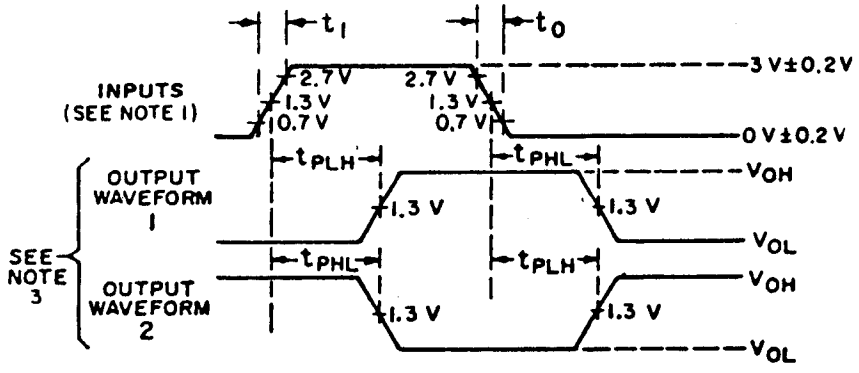
NOTES:
1. R_1 = 2 kohms, R_2 = 2 ohms.
2. V_{CC} shall be high enough to insure 5.0 V at the V_{CC} terminal. High temperature (175°C) burn-in, condition F (optional circuit) only.

FIGURE 3. Burn-in and life test circuits - Continued.

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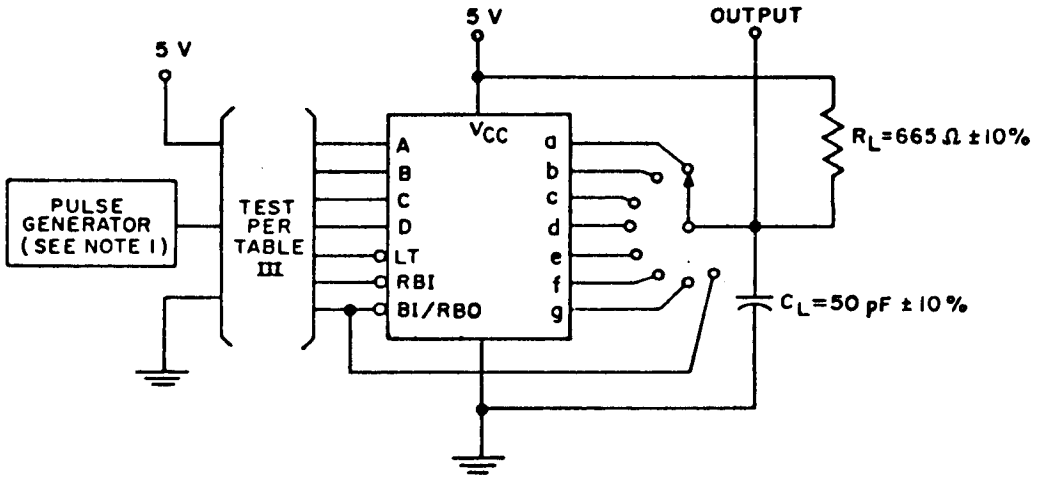
TEST CIRCUIT



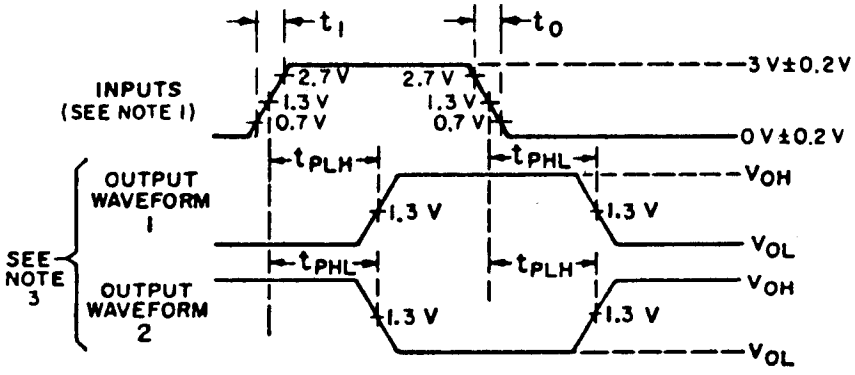
NOTES:
1. The pulse generator has the following characteristics: $t_1 \leq 15$ ns $t_0 \leq 6$ ns, PRR ≤ 1 MHz.
2. C_L includes probe and jig capacitance.
3. Input - output waveform combination in accordance with the truth tables (see figure 2).
4. All diodes are 1N3064 or equivalent.

FIGURE 4. Switching times for device types 01, 02 and 03.

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TEST CIRCUIT



NOTES:
1. Pulse generator characteristics: PRR $\leq$ 1.0 MHz, $t_1 \leq$ 15 ns, $t_0 \leq$ 6 ns, and $Z_{out} \approx 50\Omega$.
2. C_L includes probe and jig capacitance.
3. Input - output waveform combination in accordance with the truth tables (see figure 2).

FIGURE 5. Switching test circuit and waveforms for device type 04.

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TABLE III. Group A inspection for device type 01. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases E, F 1/ X, 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	A	B	C	G2A	G2B	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}				
1 T_C = +25°C	V_{OH2}	3006	1						0.7 V		GND								4.5 V	Y0	2.5		V
			2						"		"								"	Y1	"		"
			3						"		"								"	Y2	"		"
			4						"		"								"	Y3	"		"
			5						"		"								"	Y4	"		"
			6						"		"								"	Y5	"		"
			7						"		"								"	Y6	"		"
			8						"	-.4 mA	"	-.4 mA	-.4 mA	-.4 mA	-.4 mA	-.4 mA	-.4 mA	-.4 mA	"	Y7	"		"
	V_{OL2}	3007	9	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	2.0 V		"								"	Y0		0.40	"
			10	2.0 V	0.7 V	"	"	"	"		"								"	Y1		"	"
			11	0.7 V	2.0 V	"	"	"	"		"								"	Y2		"	"
			12	2.0 V	2.0 V	"	"	"	"		"								"	Y3		"	"
			13	0.7 V	0.7 V	2.0 V	"	"	"		"								"	Y4		"	"
			14	2.0 V	0.7 V	"	"	"	"		"								"	Y5		"	"
			15	0.7 V	2.0 V	"	"	"	"		"								"	Y6		"	"
			16	2.0 V	2.0 V	"	"	"	"	4 mA	"	4 mA	4 mA	4 mA	4 mA	4 mA	4 mA	4 mA	"	Y7		"	"
	V_{IC}		17	-18 mA							"								"	A		-1.5	"
			18		-18 mA						"								"	B		"	"
			19			-18 mA					"								"	C		"	"
			20				-18 mA				"								"	G2A		"	"
			21					-18 mA			"								"	G2B		"	"
			22						-18 mA		"								"	G1		"	"
	I_{IL1}	3009	23	0.4 V							"								5.5 V	A	2/	2/	2/
			24		0.4 V						"								"	B	"	"	"
			25			0.4 V					"								"	C	"	"	"
			26				0.4 V				"								"	G2A	"	"	"
			27					0.4 V			"								"	G2B	"	"	"
			28						0.4 V		"								"	G1	"	"	"
	I_{IH1}	3010	29	2.7 V							"								"	A		20	µA
			30		2.7 V						"								"	B		"	"
			31			2.7 V					"								"	C		"	"
			32				2.7 V				"								"	G2A		"	"
			33					2.7 V			"								"	G2B		"	"
			34						2.7 V		"								"	G1		"	"
	I_{IH2}	3010	35	7.0 V							"								"	A		100	"
			36		7.0 V						"								"	B		"	"
			37			7.0 V					"								"	C		"	"
			38				7.0 V				"								"	G2A		"	"
			39					7.0 V			"								"	G2B		"	"
			40						7.0 V		"								"	G1		"	"
	I_{OS}	3011	41				5.5 V	5.5 V			"								"	Y0	-15	-100	mA
			42				"	"			"						GND		"	Y1	"	"	"
			43				"	"			"					GND			"	Y2	"	"	"
			44				"	"			"				GND				"	Y3	"	"	"
			45				"	"			"			GND					"	Y4	"	"	"
			46				"	"			"		GND						"	Y5	"	"	"
			47				"	"			"	GND							"	Y6	"	"	"
			48				"	"		GND	"							GND	"	Y7	"	"	"
	I_{CC}	3005	49				GND	GND	5.5 V		"								"	V_{CC}		10.0	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = +125°C and V_{IC} tests omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests omitted.																						

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / X, 2 1/	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	A	B	C	G2A	G2B	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}				
7 3/ T_C = +25°C	Truth table test	3014	50	A	A	A	A	A	A	H	GND	H	H	H	H	H	H	H	5.0 V	3/			
			51	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			52	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			53	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"				
			54	"	"	"	B	A	"	"	"	"	"	"	"	"	"	"	"				
			55	"	"	"	A	B	"	"	"	"	"	"	"	"	"	"	"				
			56	"	"	"	A	B	B	"	"	"	"	"	"	"	"	"	"				
			57	"	"	"	B	A	"	"	"	"	"	"	"	"	"	"	"				
			58	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"				
			59	"	"	"	"	"	A	"	"	"	"	"	"	"	"	L	"				
			60	A	"	"	"	"	"	"	"	"	"	"	"	"	L	H	"				
			61	B	A	"	"	"	"	"	"	"	"	"	L	H	"	"	"				
			62	A	A	"	"	"	"	"	"	"	"	L	H	"	"	"	"				
			63	B	B	A	"	"	"	"	"	"	"	H	"	"	"	"	"				
			64	A	B	"	"	"	"	"	"	"	L	"	"	"	"	"	"				
			65	B	A	"	"	"	"	"	"	L	H	"	"	"	"	"	"				
			66	A	A	"	"	"	"	L	"	H	H	"	"	"	"	"	"				
8 3/	Truth table test	3014	67 thru 83	Same tests as subgroup 7, except T_C = +125°C.																			
	Truth table test	3014	84 thru 100	Same tests as subgroup 7, except T_C = -55°C.																			
9 T_C = +25°C	t_{PLH2} t_{PHL2}	3003 Fig. 4	101 & 102	IN	GND	GND	GND	GND	5.0 V		GND					OUT		OUT	5.0 V	A to Y0	5	25/46	ns
			103 & 104	"	5.0 V	GND	"	"	"		"								"	A to Y2	"	"	"
			105 & 106	"	GND	5.0 V	"	"	"		"			OUT					"	A to Y4	"	"	"
			107 & 108	"	5.0 V	5.0 V	"	"	"		"	OUT							"	A to Y6	"	"	"
			109 & 110	GND	IN	GND	"	"	"		"							OUT	"	B to Y0	"	"	"
			111 & 112	5.0 V	"	GND	"	"	"		"						OUT		"	B to Y1	"	"	"
			113 & 114	GND	"	5.0 V	"	"	"		"			OUT					"	B to Y4	"	"	"
			115 & 116	5.0 V	"	5.0 V	"	"	"		"		OUT						"	B to Y5	"	"	"
			117 & 118	GND	GND	IN	"	"	"		"							OUT	"	C to Y0	"	"	"
			119 & 120	5.0 V	GND	"	"	"	"		"						OUT		"	C to Y1	"	"	"
			121 & 122	GND	5.0 V	"	"	"	"		"					OUT			"	C to Y2	"	"	"
			123 & 124	5.0 V	5.0 V	"	"	"	"		"				OUT				"	C to Y3	"	"	"
	t_{PLH3} t_{PHL3}	3003 Fig. 4	125 & 126	GND	GND	GND	IN	"	"		"							OUT	"	G2A to Y0	"	23/37	"
			127 & 128	5.0 V	GND	"	"	"	"		"						OUT		"	G2A to Y1	"	"	"
			129 & 130	GND	5.0 V	"	"	"	"		"					OUT			"	G2A to Y2	"	"	"
			131 & 132	5.0 V	5.0 V	"	"	"	"		"				OUT				"	G2A to Y3	"	"	"
			133 & 134	GND	GND	5.0 V	"	"	"		"			OUT					"	G2A to Y4	"	"	"
			135 & 136	5.0 V	GND	"	"	"	"		"		OUT						"	G2A to Y5	"	"	"
			137 & 138	GND	5.0 V	"	"	"	"		"	OUT							"	G2A to Y6	"	"	"
			139 & 140	5.0 V	5.0 V	"	"	"	"	OUT	"								"	G2A to Y7	"	"	"
			141 & 142	GND	GND	GND	GND	IN	"		"							OUT	"	G2B to Y0	"	"	"
			143 & 144	5.0 V	GND	"	"	"	"		"						OUT		"	G2B to Y1	"	"	"
			145 & 146	GND	5.0 V	"	"	"	"		"					OUT			"	G2B to Y2	"	"	"
			147 & 148	5.0 V	5.0 V	"	"	"	"		"				OUT				"	G2B to Y3	"	"	"
			149 & 150	GND	GND	5.0 V	"	"	"		"			OUT					"	G2B to Y4	"	"	"
			151 & 152	5.0 V	GND	"	"	"	"		"		OUT						"	G2B to Y5	"	"	"
			153 & 154	GND	5.0 V	"	"	"	"		"	OUT							"	G2B to Y6	"	"	"
			155 & 156	5.0 V	5.0 V	"	"	"	"	OUT	"								"	G2B to Y7	"	"	"
	t_{PLH4} t_{PHL4}	3003 Fig. 4	157 & 158	IN	GND	GND	"	GND	"		"						OUT		"	A to Y1	"	32/54	"
			159 & 160	"	5.0 V	GND	"	"	"		"				OUT				"	A to Y3	"	"	"
			161 & 162	"	GND	5.0 V	"	"	"		"			OUT					"	A to Y5	"	"	"
			163 & 164	"	5.0 V	5.0 V	"	"	"		"		OUT						"	A to Y7	"	"	"
			165 & 166	GND	IN	GND	"	"	"		"					OUT			"	B to Y2	"	"	"
			167 & 168	5.0 V	"	GND	"	"	"		"				OUT				"	B to Y3	"	"	"
			169 & 170	GND	"	5.0 V	"	"	"		"	OUT							"	B to Y6	"	"	"
			171 & 172	5.0 V	"	5.0 V	"	"	"	OUT	"								"	B to Y7	"	"	"
			173 & 174	GND	GND	IN	"	"	"		"								"	C to Y4	"	"	"
			175 & 176	5.0 V	GND	"	"	"	"		"			OUT					"	C to Y5	"	"	"
			177 & 178	GND	5.0 V	"	"	"	"		"	OUT							"	C to Y6	"	"	"
			179 & 180	5.0 V	5.0 V	"	"	"	"	OUT	"								"	C to Y7	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits		Unit
			E, F																		Min	Max	
			X, 2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	A	B	C	G2A	G2B	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}				
9 T_C = +25°C	t_{PLH5} t_{PHL5}	3003 Fig. 4	181 & 182	GND	GND	GND	GND	GND	IN		GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	5.0 V	G1 to Y0	5	31/43	ns
			182 & 184	5.0 V	GND	"	"	"	"		"								"	G1 to Y1	"	"	"
			185 & 186	GND	5.0 V	"	"	"	"		"								"	G1 to Y2	"	"	"
			187 & 188	5.0 V	5.0 V	"	"	"	"		"								"	G1 to Y3	"	"	"
			189 & 190	GND	GND	5.0 V	"	"	"		"								"	G1 to Y4	"	"	"
			191 & 192	5.0 V	GND	"	"	"	"		"								"	G1 to Y5	"	"	"
			193 & 194	GND	5.0 V	"	"	"	"		"								"	G1 to Y6	"	"	"
			195 & 196	5.0 V	5.0 V	"	"	"	"		"								"	G1 to Y7	"	"	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and for following limits: t_{PLH2} = 5 to 38 ns; t_{PHL2} = 5 to 69 ns; t_{PLH3} = 5 to 35 ns; t_{PHL3} = 5 to 56 ns; t_{PLH4} = 5 to 48 ns; t_{PHL4} = 5 to 81 ns; t_{PLH5} = 5 to 47 ns; t_{PHL5} = 5 to 65 ns.																						
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																						

1/ Pins not designated are high ≥2.0 V; low ≤0.7 V; or open. Case 2 pins not referenced are NC.

2/ I_{IL1} limits are as follows:

Test	Min/Max limits (mA) for circuits				
	A	B	C, E, F	D	G
I_{IL1}	-.001/-.150 except -.11/-.35 for tests 26, 27, 28	-.030/-.300	-.12/-.36	-.10/-.34	-.15/-.38

3/ A = 2.5 V and B = 0.4 V; H ≥1.5 V; L ≤1.5 V;

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TABLE III. Group A inspection for device type 02. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases E, F 1/ X, Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	1G	1A	1B	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	2B	2A	2G	V_{CC}				
T_C = 1/ +25°C	V_{OH2}	3006	1	2.0 V							GND								4.5 V	1Y3	2.5		V
			2	"							"								"	1Y2	"		"
			3	"							"								"	1Y1	"		"
			4				-.4 mA				"								"	1Y0	"		"
			5					-.4 mA			"								"	2Y3	"		"
			6						-.4 mA		"								"	2Y2	"		"
			7							-.4 mA	"								"	2Y1	"		"
			8								"	-.4 mA	-.4 mA	-.4 mA	-.4 mA			2.0 V	"	2Y0	"		"
	V_{OL2}	3007	9	0.7 V	2.0 V	2.0 V					"								"	1Y3		0.4	"
			10	"	0.7 V	2.0 V					"								"	1Y2		"	"
			11	"	2.0 V	0.7 V					"								"	1Y1		"	"
			12		0.7 V	0.7 V	4 mA				"								"	1Y0		"	"
			13					4 mA			"								"	2Y3		"	"
			14						4 mA		"								"	2Y2		"	"
			15							4 mA	"								"	2Y1		"	"
			16								"	4 mA	4 mA	4 mA	4 mA	2.0 V	2.0 V	0.7 V	"	2Y0		"	"
																2.0 V	0.7 V	"					
																0.7 V	2.0 V	"					
																0.7 V	0.7 V	"					
	V_{IC}		17	-18 mA	-18 mA						"								"	1G		-1.5	"
			18			-18 mA					"								"	1A		"	"
			19								"								"	1B		"	"
			20								"					-18 mA	-18 mA		"	2B		"	"
			21								"								"	2A		"	"
			22								"								"	2G		"	"
	I_{IL1}	3009	23	0.4 V							"								5.5 V	1G	2/	2/	mA
			24	"	0.4 V						"								"	1A	"	"	"
			25	"	"	0.4 V					"								"	1B	"	"	"
			26		"	"					"					0.4 V			"	2B	"	"	"
			27			"					"					"	0.4 V		"	2A	"	"	"
			28								"						"	0.4 V	"	2G	"	"	"
	I_{IH1}	3010	29	2.7 V							"								"	1G		20	µA
			30	"	2.7 V						"								"	1A		"	"
			31	"	"	2.7 V					"								"	1B		"	"
			32		"	"					"					2.7 V			"	2B		"	"
			33			"					"					"	2.7 V		"	2A		"	"
			34								"					"	"	2.7 V	"	2G		"	"
	I_{IH2}	3010	35	7.0 V							"								"	1G		100	µA
			36	"	7.0 V						"								"	1A		"	"
			37	"	"	7.0 V					"								"	1B		"	"
			38		"	"					"					7.0 V			"	2B		"	"
			39			"					"					"	7.0 V		"	2A		"	"
			40								"					"	"	7.0 V	"	2G		"	"
	I_{OS}	3011	41	5.5 V			GND				"								"	1Y0	-15	-100	mA
			42	"				GND			"								"	1Y1	"	"	"
			43	"					GND		"								"	1Y2	"	"	"
			44							GND	"								"	1Y3	"	"	"
			45								"	GND							"	2Y3	"	"	"
			46								"		GND						"	2Y2	"	"	"
			47								"			GND					"	2Y1	"	"	"
			48								"				GND				"	2Y0	"	"	"
	I_{CC}	3005	49	GND	5.5 V	5.5 V					"					5.5 V	5.5 V	GND	"	V_{CC}		11.0	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = +125°C and V_{IC} tests omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests omitted.																						

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / X, 2 1/ / Test no.	1 / 2 / 1G	2 / 3 / 1A	3 / 4 / 1B	4 / 5 / 1Y0	5 / 7 / 1Y1	6 / 8 / 1Y2	7 / 9 / 1Y3	8 / 10 / GND	9 / 12 / 2Y3	10 / 13 / 2Y2	11 / 14 / 2Y1	12 / 15 / 2Y0	13 / 17 / 2B	14 / 18 / 2A	15 / 19 / 2G	16 / 20 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
7 3/ T_C = +25°C	Truth table test	3014	50	A	A	A	H	H	H	H	GND	H	H	H	H	A	A	A	5.0 V	3/			
			51	"	B	A	"	"	"	"	"	"	"	"	"	A	B	"	"				
			52	"	A	B	"	"	"	"	"	"	"	"	"	B	A	"	"				
			53	"	B	B	"	"	"	"	"	"	"	"	"	B	B	"	"				
			54	B	B	A	L	L	L	"	"	"	L	L	L	A	A	B	"				
			55	"	A	B	H	H	H	"	"	"	H	H	"	A	B	"	"				
			56	"	B	A	"	H	L	"	"	"	H	H	"	B	A	"	"				
			57	"	A	A	"	H	H	L	"	L	H	H	"	A	A	"	"				
8 3/	Truth table test	3014	58 thru 65	Same tests as subgroup 7, except T_C = +125°C.																			
	Truth table test	3014	66 thru 73	Same tests as subgroup 7, except T_C = -55°C.																			
9 T_C = +25°C	t_{PLH3} t_{PHL3}	3003 Fig. 4	74 & 75	IN	GND	GND	OUT	OUT	OUT	OUT	GND								5.0 V	1G to 1Y0	5	29/37	ns
			76 & 77	"	5.0 V	GND					"								"	1G to 1Y1	"	"	"
			78 & 79	"	GND	5.0 V					"								"	1G to 1Y2	"	"	"
			80 & 81	"	5.0 V	5.0 V					"								"	1G to 1Y3	"	"	"
	t_{PLH2} t_{PHL2}	3003 Fig. 4	82 & 83	GND	IN	GND	OUT		OUT										"	1A to 1Y0	"	25/38	"
			84 & 85	"	IN	5.0 V	OUT				"								"	1A to 1Y2	"	"	"
			86 & 87	"	GND	IN		OUT			"								"	1B to 1Y0	"	"	"
			88 & 89	"	5.0 V	IN					"								"	1B to 1Y1	"	"	"
	t_{PLH3} t_{PHL3}	3003 Fig. 4	90 & 91																"	2G to 2Y0	"	29/37	"
			92 & 93																"	2G to 2Y1	"	"	"
			94 & 95																"	2G to 2Y2	"	"	"
			96 & 97																"	2G to 2Y3	"	"	"
	t_{PLH2} t_{PHL2}	3003 Fig. 4	98 & 99																"	2A to 2Y0	"	25/38	"
			100 & 101																"	2A to 2Y2	"	"	"
			102 & 103																"	2B to Y20	"	"	"
			104 & 105																"	2B to 2Y1	"	"	"
	t_{PLH4} t_{PHL4}	3003 Fig. 4	106 & 107	GND	IN	GND		OUT		OUT									"	1A to 1Y1	"	34/43	"
			108 & 109	"	IN	5.0 V				OUT	"								"	1A to 1Y3	"	"	"
			110 & 111	"	GND	IN					"								"	1B to 1Y2	"	"	"
			112 & 113	"	5.0 V	IN					"								"	1B to 1Y3	"	"	"
			114 & 115																"	2A to 2Y1	"	"	"
			116 & 117																"	2A to 2Y3	"	"	"
			118 & 119																"	2B to 2Y2	"	"	"
			120 & 121																"	2B to 2Y3	"	"	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and for following limits: t_{PLH4} = 5 to 51 ns; t_{PHL4} = 5 to 65 ns.																						
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																						

1/ Pins not designated are high ≥2.0 V; low ≤0.7 V; or open. Case 2 pins not referenced are NC.

2/ I_{IL1} limits are as follows:

Test	Min/Max limits (mA) for circuits			
	A, C, E, F	B	D	G
I_{IL1}	-.12/-.36	-.03/-.300	-.10/-.34	-.15/-.38

3/ H ≥1.5 V; L ≤1.5 V; A = 2.5 V; B = 0.4 V.

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TABLE III. Group A inspection for device type 03. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
			X, Z 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}				
T_C = +25°C	V_{OH2}	3006	1	-.4 mA							GND				2.0 V	2.0 V	2.0 V	2.0 V	4.5 V	0	2.5		V
			2		-.4 mA						"				"	"	"	"	"	1	"		"
			3			-.4 mA					"				"	"	"	"	"	2	"		"
			4				-.4 mA				"				"	"	"	"	"	3	"		"
			5					-.4 mA			"				"	"	"	"	"	4	"		"
			6						-.4 mA		"				"	"	"	"	"	5	"		"
			7							-.4 mA	"				"	"	"	"	"	6	"		"
			8								"	-.4 mA			"	"	"	"	"	7	"		"
			9								"		-.4 mA		"	"	"	"	"	8	"		"
			10								"			-.4 mA	"	"	"	"	"	9	"		"
	V_{OL2}	3007	11	4 mA							"				0.7 V	0.7 V	0.7 V	0.7 V	"	0		0.4	"
			12		4 mA						"				"	"	2.0 V	2.0 V	"	1		"	"
			13			4 mA					"				"	"	0.7 V	0.7 V	"	2		"	"
			14				4 mA				"				"	"	2.0 V	2.0 V	"	3		"	"
			15					4 mA			"				"	2.0 V	0.7 V	0.7 V	"	4		"	"
			16						4 mA		"				"	"	0.7 V	2.0 V	"	5		"	"
			17							4 mA	"				"	"	2.0 V	0.7 V	"	6		"	"
			18								"	4 mA			"	"	0.7 V	2.0 V	"	7		"	"
			19								"		4 mA		"	"	2.0 V	0.7 V	"	8		"	"
			20								"				2.0 V	0.7 V	0.7 V	2.0 V	"	9		"	"
	V_{IC}		21								"				-18 mA				"	A		-1.5	"
			22								"				"	-18 mA			"	B		"	"
			23								"				"	"	-18 mA		"	C		"	"
			24								"				"	"	"	-18 mA	"	D		"	"
	I_{IH1}	3010	25								"				2.7 V				5.5 V	A		20	µA
			26								"				"	2.7 V			"	B		"	"
			27								"				"	"	2.7 V	2.7 V	"	C		"	"
			28								"				"	"	"	"	"	D		"	"
	I_{IH2}	3010	29								"				7.0 V				"	A		100	"
			30								"				"	7.0 V			"	B		"	"
			31								"				"	"	7.0 V	7.0 V	"	C		"	"
			32								"				"	"	"	"	"	D		"	"
	I_{IL1} 2/	3009	33								"				0.4 V				"	A	2/	2/	mA
			34								"				"	0.4 V			"	B	"	"	"
			35								"				"	"	0.4 V	0.4 V	"	C	"	"	"
			36								"				"	"	"	"	"	D	"	"	"
	I_{OS} 3/	3011	37	GND							"				5.5 V				"	0	-15	-100	"
			38		GND						"				"	5.5 V			"	1	"	"	"
			39			GND					"				"	"	5.5 V	5.5 V	"	2	"	"	"
			40				GND				"				"	"	"	"	"	3	"	"	"
			41					GND			"				"	"	"	"	"	4	"	"	"
			42						GND		"				"	"	"	"	"	5	"	"	"
			43							GND	"				"	"	"	"	"	6	"	"	"
			44								"	GND			"	"	"	"	"	7	"	"	"
			45								"		GND		"	"	"	"	"	8	"	"	"
			46								"			GND	"	"	"	"	"	9	"	"	"
	I_{CC}	3005	47								"				GND	GND	GND	GND	"	V_{CC}		13	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = +125°C and V_{IC} tests omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests omitted.																						

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
			X, 2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}				
7 4/ T_C = +25°C	Truth table test	3014	48	L	H	H	H	H	H	H	GND	H	H	H	3	3	3	3	5.0 V	4/			
			49	H	L	H	H	"	"	"	"	"	"	"	"	"	3	A	"				
			50	"	H	L	"	"	"	"	"	"	"	"	"	"	A	3	"				
			51	"	"	H	L	L	"	"	"	"	"	"	"	A	3	A	"				
			52	"	"	"	H	H	"	"	"	"	"	"	"	"	B	A	"				
			53	"	"	"	"	L	L	"	"	"	"	"	"	"	A	A	"				
			54	"	"	"	"	H	H	"	"	"	"	"	"	"	A	A	"				
			55	"	"	"	"	"	"	L	"	"	"	"	A	B	B	3	"				
			56	"	"	"	"	"	"	H	"	"	"	"	"	"	B	3	"				
			57	"	"	"	"	"	"	"	"	"	"	"	"	"	A	3	"				
			58	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"				
			59	"	"	"	"	"	"	"	"	"	"	"	"	"	B	A	"				
			60	"	"	"	"	"	"	"	"	"	"	"	"	"	B	A	"				
			61	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"				
			62	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"				
			63	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"				
8 4/	Truth table test	3014	64 thru 69	Same tests as subgroup 7, except T_C = +125°C.																			
	Truth table test	3014	80 thru 95	Same tests as subgroup 7, except T_C = -55°C.																			
9 T_C = +25°C	t_{PHL4} t_{PLH4}	3003 Fig. 4	96 & 97		OUT	OUT					GND	OUT			GND	GND	IN	GND	5.0 V	B to 2	5	35	ns
			98 & 99								"				"	5.0 V	GND	IN	"	A to 1	"	"	"
			100 & 101								"				"	"	"	IN	"	A to 7	"	"	"
	t_{PHL2} t_{PLH2}	3003 Fig. 4	102 & 103	OUT	OUT			OUT			"				"	GND	IN	5.0 V	"	B to 1	"	30	"
			104 & 105			OUT	OUT				"				"	5.0 V	GND	GND	"	B to 4	"	"	"
			106 & 107								"				"	GND	IN	GND	"	A to 0	"	"	"
			108 & 109								"				"	IN	5.0 V	5.0 V	"	C to 2	"	"	"
			110 & 111								"				"	"	5.0 V	5.0 V	"	C to 3	"	"	"
			112 & 113								"				"	"	GND	GND	"	C to 8	"	"	"
			114 & 115								"				5.0 V	"	"	5.0 V	"	C to 9	"	"	"
	t_{PHL4} t_{PLH4}	3003 Fig. 4	116 & 117					OUT	OUT	OUT	"	OUT			GND	"	"	GND	"	C to 4	"	35	"
			118 & 119								"				"	"	"	5.0 V	"	C to 5	"	"	"
			120 & 121								"				"	"	5.0 V	GND	"	C to 6	"	"	"
			122 & 123								"				"	"	"	5.0 V	"	C to 7	"	"	"
	t_{PHL2} t_{PLH2}	3003 Fig. 4	124 & 125				OUT	OUT	OUT	OUT	"	OUT			IN	GND	"	5.0 V	"	D to 3	"	30	"
			126 & 127								"				"	5.0 V	GND	GND	"	D to 4	"	"	"
			128 & 129								"				"	"	GND	GND	"	D to 5	"	"	"
			130 & 131								"				"	"	5.0 V	GND	"	D to 6	"	"	"
			132 & 133								"				"	"	5.0 V	5.0 V	"	D to 7	"	"	"
	t_{PHL4} t_{PLH4}	3003 Fig. 4	134 & 135								"		OUT	OUT	"	GND	GND	GND	"	D to 8	"	35	"
			136 & 137								"				"	GND	GND	5.0 V	"	D to 9	"	35	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and for following limits: t_{PLH2} = 5 to 45 ns; t_{PHL2} = 5 to 45 ns; t_{PLH4} = 5 to 53 ns; t_{PHL4} = 5 to 53 ns.																						
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																						

1/ Pins not designated are high ≥2.0 V; low ≤0.7 V; or open. Case 2 pins not referenced are NC.

2/ I_{IL1} limits shall be as follows:

Min/Max limits (mA) for circuits						
A	B	C	D	E	F	G
-.12/-.36	-.030/-.300	-.12/-.36	-.12/-.36	-.12/-.36	-.12/-.36	-.16/-.40

3/ For circuit C, limits are: -15 to -130 mA.

4/ H ≥1.5 V; L ≤1.5 V; A = 2.5 V; B = 0.4 V.

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TABLE III. Group A inspection for device type 04. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / X, Z 1/	1 / 2	2 / 3	3 / 4	4 / 5	5 / 7	6 / 8	7 / 9	8 / 10	9 / 12	10 / 13	11 / 14	12 / 15	13 / 17	14 / 18	15 / 19	16 / 20	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	B	C	LT	RBO	RBI	D	A	GND	e	d	c	b	a	g	f	V_{CC}				
1 T_C = +25°C	V_{OL2}	3006	1	X	X	0.7 V		X	X	X	GND	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	12 mA	4.5 V	e		0.4	V
			2	"	"	"		"	"	"	"								"	d		"	"
			3	"	"	"		"	"	"	"								"	c		"	"
			4	"	"	"		"	"	"	"								"	b		"	"
			5	"	"	"		"	"	"	"								"	a		"	"
			6	"	"	"		"	"	"	"								"	g		"	"
			7	"	"	"		"	"	"	"								"	F		"	"
	V_{OL1}	3006	8	0.7 V	0.7 V	2.0 V	1.6 mA	0.7 V	0.7 V	0.7 V	"								"	RBO		"	"
	I_{CEX}		9	"	"	"		"	"	"	"	15 V	15 V	15 V	15 V	15 V	15 V	15 V	"	e		250	µA
			10	"	"	"		"	"	"	"								"	d		"	"
			11	"	"	"		"	"	"	"								"	c		"	"
			12	"	"	"		"	"	"	"								"	b		"	"
			13	"	"	"		"	"	"	"								"	a		"	"
			14	"	"	"		"	"	"	"								"	g		"	"
			15	"	"	"		"	"	"	"								"	f		"	"
	V_{OH1}	3007	16	0.7 V	0.7 V	2.0 V	-50 µA	2.0 V	0.7 V	0.7 V	"								"	RBO	2.4		V
	V_{IC}		17	-18 mA							"								"	B		-1.5	"
			18		-18 mA						"								"	C		"	"
			19			-18 mA					"								"	LT		"	"
			20					-18 mA			"								"	RBI		"	"
			21						-18 mA		"								"	D		"	"
			22							-18 mA	"								"	A		"	"
	I_{IL1}	3009	23	0.4 V	5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	"								5.5 V	B	-.10	-.34	mA
			24	5.5 V	0.4 V	"		"	5.5 V	"	"								"	C	"	"	"
			25	"	5.5 V	"		"	0.4 V	"	"								"	D	"	"	"
			26	"	"	"		"	5.5 V	0.4 V	"								"	A	"	"	"
	I_{IL2}	3009	27	"	"	0.4 V		"	"	5.5 V	"								"	LT	2/	2/	"
	I_{IL2}		28	"	"	5.5 V		0.4 V	"	"	"								"	RBI	"	"	"
	I_{IL3}	3009	29	"	"	5.5 V	0.4 V	5.5 V	"	"	"								"	RBO	"	"	"
	I_{IH1}	3010	30	2.7 V	GND	GND		GND	GND	GND	"								"	B		20	µA
			31	GND	2.7 V	GND		"	"	"	"								"	C		"	"
			32	"	GND	2.7 V		"	"	"	"								"	LT		"	"
			33	"	"	GND		2.7 V	"	"	"								"	RBI		"	"
			34	"	"	"		GND	2.7 V	"	"								"	D		"	"
			36	"	"	"		"	GND	2.7 V	"								"	A		"	"
	I_{IH2}	3010	36	7.0 V	"	"		"	"	GND	"								"	B		100	"
			37	GND	7.0 V	"		"	"	"	"								"	C		"	"
			38	"	GND	7.0 V		"	"	"	"								"	LT		"	"
			39	"	"	GND		7.0 V	"	"	"								"	RBI		"	"
			40	"	"	"		GND	7.0 V	"	"								"	D		"	"
			41	"	"	"		GND	GND	7.0 V	"								"	A		"	"
	I_{CC}	3005	42	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"								"	V_{CC}		13	mA
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = +125°C and V_{IC} tests omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests omitted.																						

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
			Cases X, 2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B	C	LT	RBO	RBI	D	A	GND	e	d	c	b	a	g	f	V_{CC}				
7 T_C = +25°C 3/ 4/	Truth table test	3014	43	B	B	A	A	A	B	B	GND	L	L	L	L	L	H	L	5.0 V	5/			
			44	B	"	"	"	X	"	A	"	H	H	L	"	L	H	H	"				
			45	A	"	"	"	"	"	B	"	L	L	H	"	L	L	L	"				
			46	A	"	"	"	"	"	A	"	H	L	L	"	H	L	H	"				
			47	B	A	"	"	"	"	B	"	"	H	"	"	L	H	"	"				
			48	B	"	"	"	"	"	A	"	L	L	"	"	H	L	L	"				
			49	A	"	"	"	"	"	B	"	H	H	"	"	H	"	H	"				
			50	A	"	"	"	"	"	A	"	L	L	"	"	L	"	L	"				
			51	B	B	"	"	"	A	B	"	H	H	"	"	H	"	H	"				
			52	B	"	"	"	"	"	A	"	L	L	"	"	L	"	L	"				
			53	A	"	"	"	"	"	B	"	H	H	"	"	H	"	H	"				
			54	A	"	"	"	"	"	A	"	L	L	"	"	L	"	L	"				
			55	B	A	"	"	"	"	B	"	H	H	"	"	H	"	H	"				
			56	B	"	"	"	"	"	A	"	L	L	"	"	L	"	L	"				
			57	A	"	"	"	"	"	B	"	H	H	"	"	H	"	H	"				
			58	A	"	"	"	"	"	A	"	L	L	"	"	L	"	L	"				
			59	X	X	X	"	"	X	X	"												
			60	B	B	5.0 V	L	B	B	B	"												
			61	X	X	8	A	X	X	X	"												
8 3/ 4/	Truth table test	3014	62 thru 80	Same tests as subgroup 7, except T_C = +125°C.																			
	Truth table test	3014	81 thru 99	Same tests as subgroup 7, except T_C = -55°C.																			
9 T_C = +25°C	t_{PHL1} t_{PLH1}	3003 Fig. 5	100 & 101	GND	GND	5.0 V		5.0 V	GND	IN	GND					OUT			5.0 V	A to a	5	105	ns
			102 & 103	GND	5.0 V	"		"	5.0 V	"	"					"			"	A to a	"	"	"
			104 & 105	5.0 V	"	"		"	GND	"	"				OUT				"	A to b	"	"	"
			106 & 107	GND	"	"		"	"	"	"				OUT				"	A to b	"	"	"
			108 & 109	5.0 V	"	"		"	"	"	"								"	A to c	"	"	"
			110 & 111	5.0 V	GND	"		"	"	"	"								"	A to d	"	"	"
			112 & 113	GND	GND	"		"	"	"	"								"	A to d	"	"	"
			114 & 115	GND	5.0 V	"		"	"	"	"								"	A to d	"	"	"
			116 & 117	5.0 V	5.0 V	"		"	"	"	"								"	A to e	"	"	"
			118 & 119	GND	GND	"		"	"	"	"								"	A to e	"	"	"
			120 & 121	5.0 V	"	"		"	"	"	"							OUT	"	A to f	"	"	"
			122 & 123	GND	"	"		"	"	"	"							OUT	"	A to f	"	"	"
			124 & 125	5.0 V	5.0 V	"		"	5.0 V	"	"								"	A to g	"	"	"
			126 & 127	5.0 V	5.0 V	"		"	5.0 V	"	"						OUT		"	A to RBO	"	"	"
			128 & 129	GND	GND	"		GND	GND	GND	"					OUT			"	B to a	"	"	"
			130 & 131	IN	"	"		5.0 V	"	"	"					OUT			"	B to a	"	"	"
			132 & 133	"	"	"		"	GND	5.0 V	"								"	B to b	"	"	"
			134 & 135	"	5.0 V	"		"	GND	GND	"				OUT				"	B to b	"	"	"
			136 & 137	"	GND	"		"	5.0 V	GND	"				"				"	B to b	"	"	"
			138 & 139	"	5.0 V	"		"	GND	5.0 V	"				"				"	B to b	"	"	"
			140 & 141	"	GND	"		"	5.0 V	5.0 V	"			OUT					"	B to c	"	"	"
			142 & 143	"	"	"		"	GND	GND	"								"	B to d	"	"	"
			144 & 145	"	"	"		"	"	5.0 V	"			OUT					"	B to d	"	"	"
			146 & 147	"	5.0 V	"		"	"	GND	"			OUT					"	B to e	"	"	"
			148 & 149	"	"	"		"	"	GND	"								"	B to f	"	"	"
			150 & 151	"	"	"		"	"	5.0 V	"							OUT	"	B to f	"	"	"
			152 & 153	"	GND	"		"	"	GND	"								"	B to g	"	"	"
			154 & 155	"	"	"		"	5.0 V	"	"					OUT			"	C to a	"	"	"
			156 & 157	"	"	"		"	GND	"	"								"	C to b	"	"	"
			158 & 159	GND	IN	"		"	5.0 V	"	"				OUT				"	C to b	"	"	"
			160 & 161	5.0 V	"	"		"	GND	"	"								"	C to c	"	"	"
			162 & 163	GND	"	"		"	5.0 V	5.0 V	"			OUT					"	C to d	"	"	"
			164 & 165	"	"	"		"	GND	GND	"								"	C to e	"	"	"
			166 & 167	"	"	"		"	GND	"	"								"	C to f	"	"	"
			168 & 169	5.0 V	"	"		"	5.0 V	"	"							OUT	"		"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
			X, 2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	B	C	LT	RBO	RBI	D	A	GND	e	d	c	b	a	g	f	V_{CC}				
9 T_C = +25°C	t_{PHL1} t_{PLH1}	3003 Fig. 5	170 & 171	GND	GND	IN		5.0 V	GND	5.0 V	GND									C to f	5	105	ns
			172 & 173	5.0 V	5.0 V	"		"	IN	5.0 V	"					OUT				D to a	"	"	"
			174 & 175	GND	5.0 V	"		"	"	GND	"			OUT						D to c	"	"	"
			176 & 177	"	GND	"		"	"	"	"						OUT			D to g	"	"	"
			178 & 179	"	"	"		GND	GND	"	"					OUT				LT to a	"	"	"
			180 & 181	"	"	"	OUT	GND	"	"	"									LT to RBO	"	"	"
			182 & 183	"	"	5.0 V	IN	5.0 V	"	"	"					OUT				RBO to a	"	"	"
			184 & 185	"	"	"	"	IN	"	"	"					OUT				RBI to a	"	"	"
			186 & 187	"	"	"	OUT	IN	"	"	"									RBI to RBO	"	"	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and for following limits:											t_{PHL1} = 5 to 158 ns; t_{PLH1} = 5 to 158 ns.											
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																						

1/ Pins not designated are high ≥2.0 V; low ≤0.7 V; or open. Case 2 pins not referenced are NC.

2/ Test limits shall be as follows:

Test	Min/Max limits (mA) for circuits A	E	G
I_{IL2}	-.12/-.36	-.11/-.35	-.12/-.36
I_{IL3}	-.36/-1.08	-.36/-1.37	-.36/-1.08

3/ X = Input may be high level or low level.

4/ A pullup resistor of 665Ω to 2 kΩ shall be connected between each output and V_{CC}.

5/ H ≥1.5 V; L ≤1.5 V; A = 2.5 V; B = 0.4 V.

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II, herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, E, or F, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours except, as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for original equipment design application and logistic support of existing equipment.

6.2 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to acquiring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

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f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirement for JAN marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - Ground, zero voltage potential.

V_{IN} - - - - - - - - - - - - Voltage level at an input terminal.

V_{IC} - - - - - - - - - - - - Input clamp voltage.

RBI - - - - - - - - - - - - - Ripple blanking input.

RBO - - - - - - - - - - - - - Ripple blanking output.

BI- - - - - - - - - - - - - - Blanking input.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed Generic-Industry type. Generic-Industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of Generic-Industry types for MIL-M-38510 types or as a waiver of any of tne provisions of MIL-M-38510.

Military device type	Generic-Industry type
01	54LS138
02	54LS139
03	54LS42
04	54LS47

6.6 Manufacturer's designation. Manufacturer circuits included in this specification are designated as shown in table IV herein.

TABLE IV. Manufacturer's designator.

Device type	A Texas Instruments	B Signetics	C Raytheon	D Fairchild	E Motorola	F Advanced Micro Devices	G National Semiconductor
01	X	X	X	X	X	X	X
02	X	X	X	X	X	X	X
03	X	X		X	X		X
04	X				X		X

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
Navy - OS, SH, TD
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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